

Savitribai Phule Pune University, Pune

Maharashtra, India



Faculty of Science and Technology



National Education Policy (NEP)-2020 Compliant Curriculum

TE - Third Year Engineering (2024 Pattern)

in

Mechanical Engineering

(With effect from Academic Year 2026-27)

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Nomenclature

AEC	Ability Enhancement Courses
AICTE	All India Council for Technical Education
CO	Course Outcome
CEP	Community Engagement Project
CCE	Comprehensive Continuous Evaluation
HSSM	Humanities, Social Science, and Management
MDM	Multidisciplinary Minor
MEC	Mechanical Engineering
MOOC	Massive Open Online Course
NPTEL	National Programme on Technology Enhanced Learning
NEP	National Education Policy
PCC	Program Core Course
PEO	Program Educational Objectives
PSO	Program Specific Objectives
SWAYAM	Study Webs of Active-learning for Young Aspiring Minds
UGC	University Grants Commission
VEC	Value Education Course
VSE	Vocational and Skill Enhancement Course
WK	Knowledge and Attitude Profile

Preface by Board of Studies

Dear Students and Teachers,

We, the members of the Board of Studies – Mechanical Engineering, are very happy to present the Third Year Mechanical Engineering syllabus, effective from the **Academic Year 2026–27 (2024 Pattern)**. We are confident that you will find this syllabus both interesting and challenging. The present curriculum will be implemented for Third Year Engineering from the academic year 2026–27, and it will be subsequently extended to the Final Years in the academic years **2027–28**.

Mechanical Engineering is one of the most sought-after branches among engineering students, which necessitates continuous revision and up gradation of the syllabus. Mechanical Engineering is a dynamic discipline that integrates principles from core engineering fields and supports innovation across manufacturing, design, energy, materials, and automation. This curriculum is designed to provide students with a comprehensive understanding of the fundamentals, emerging technologies, and practical applications in Mechanical Engineering, while also equipping them to meet the demands of a rapidly evolving industry.

The revised syllabus aligns with the vision of **NEP-2020**, and conforms to the frameworks set by Savitribai Phule Pune University, AICTE New Delhi, UGC, and various accreditation agencies. It takes into account recent technological developments, innovations, and industry needs to ensure students are well- prepared for professional challenges.

Wherever applicable, additional learning resources such as NPTEL and SWAYAM links are provided at the end of each course. Students are encouraged to utilize these platforms for self-learning, engage in online courses, and undertake additional projects to enhance their knowledge and skill set. On successful completion, they are advised to submit their course certifications, which will further support and enrich their academic growth.

This curriculum is the result of collaborative efforts involving academic experts, industry professionals, and alumni to ensure relevance and excellence. It is designed not only to meet current industry expectations but also to prepare students for higher studies, research, and entrepreneurial ventures in the field of Mechanical Engineering.

We hope this curriculum inspires students to become technically competent professionals, responsible citizens, and contributors to the technological and sustainable advancement of society.



Dr. Pradeep A. Patil
Chairman
Board of Studies - Mechanical Engineering

Program Specific Outcomes

- PSO1** : **SPECIFY, DESIGN** and **EVALUATE** mechanical components and systems using modelling and analysis software.
- PSO2** : **APPLY** knowledge of machines, tools, automation, properties of advanced materials and modern management methods for manufacturing of mechanical components and systems.
- PSO3** : **APPLY** core aspects of thermal and fluid engineering to determine the performance of mechanical systems including power absorbing and power generating systems.

Program Educational Objectives

Program Educational Objectives (PEOs) are broad statements that describe the career and professional accomplishments that the program is preparing a graduate to achieve.

- PEO1** : The graduate will have a successful career in mechanical engineering with strong technical, research & professional skills.
- PEO2** : The graduate will possess an ability to work in diversified fields along with team work and leadership qualities.
- PEO3** : The graduate will continue to learn and to adapt in a society of constantly evolving technological environment

Program Outcomes

Program Outcomes (POs) are statements that articulate what students are expected to know, understand, and be able to do by the time they graduate from the program. These outcomes are aligned with the overall educational objectives of the program and reflect the skills, knowledge, attitudes, and behaviors acquired by students throughout their academic journey. On successful completion of B.E. in Mechanical Engineering, graduating students/graduates will be able to:

PO No.	Title	Program Outcome Description
PO1	Engineering Knowledge	Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems.
PO2	Problem Analysis	Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	Design / Development of Solutions	Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment. (WK5)
PO4	Conduct Investigations of Complex Problems	Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8)
PO5	Engineering Tool Usage	Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling, recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6	The Engineer and The World	Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)
PO7	Ethics	Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8	Individual and Collaborative Team Work	Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	Communication	Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
PO10	Project Management and Finance	Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.
PO11	Life-Long Learning	Recognize the need for, and have the preparation and ability for: (i) independent and life-long learning, (ii) adaptability to new and emerging technologies, and (iii) critical thinking in the broadest context of technological change. (WK8)

Knowledge and Attitude Profile (WK)

WK No.	Focus Area	Description
WK1	Natural Sciences and Social Sciences	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Mathematics and Data Analysis	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	Engineering Fundamentals	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	Engineering Specialist Knowledge	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	Engineering Design and Environmental Considerations	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	Engineering Practice (Technology)	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK7	Role of Engineering in Society	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8	Research and Critical Thinking	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9	Ethics and Inclusive Behavior	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability, etc., with mutual respect.

Reference: Self-Assessment Report (SAR) Format Undergraduate Engineering Programs Graduate Attributes and Professional Competencies Version 4.0 (GAPC V4.0) - (August 2024) Page 55-56

General Rules and Guidelines

Term	Definition
Course Outcomes (COs)	Course Outcomes are narrower statements that describe what students are expected to know and be able to do at the end of each course. These relate to the skills, knowledge, and behavior that students acquire throughout the course.
Assessment	Assessment is one or more processes, carried out by the institution, that identify, collect, and prepare data to evaluate the achievement of Program Educational Objectives (PEOs) and Program Outcomes (POs) .
Evaluation	Evaluation is one or more processes, performed by the Evaluation Team , to interpret the data and evidence gathered through assessment practices. It determines how well PEOs or POs are being achieved, and informs decisions for improvement.

Assessment and Evaluation:

Assessment and Evaluation shall be conducted in two parts:

1. Comprehensive Continuous Evaluation (CCE)
2. End-Semester Examination (ESE)

Component	Description	Marks
Comprehensive Continuous Evaluation (CCE)	Conducted at institute level, covering all Units of the syllabus. The design and mark allocation follow the Continuous Assessment Sheet structure.	30
End-Semester Examination (ESE)	Conducted at university level, typically covering the entire syllabus through summative examination.	70
Total Marks per Subject		100

A) Comprehensive Continuous Evaluation (CCE) :

CCE to be conducted either on the basis of Mode 1 or Mode 2.

➤ CCE MODE 1:

To design a Comprehensive Continuous Evaluation (CCE) scheme for a theory subject of 15 or 30 marks with the specified parameters, the allocation of marks and the structure can be as per continuous assessment sheet;

Savitribai Phule Pune University																							
Board of Studies (Mechanical and Automobile Engineering)																							
Comprehensive Continous Evaluation (CCE) 30 Marks Distribution																							
Class: TE-Mechanical Engineering												Subject: Refrigeration and Air conditioning											
Exam Seat No.	Roll No.	Name of Student											Cumulative Sum				30 Marks Distribution				Marks obtained out		
			Unit 1		Unit 2		Unit 3		Unit 4		Unit 5		Field Activity		Quiz		Internal Test	Attendance	Field Activity			Internal Test	
			Field Activity	Quiz	Field Activity	Quiz	Field Activity	Quiz	Field Activity	Quiz	Field Activity	Quiz											
			A	B	C	D	E	F	G	H	I	J	SUM(A+C+E+G+I)	SUM(B+D+F+H+J)			P	Q	R	S		SUM(P:Q)	
			10	10	10	10	10	10	10	10	10	10	50	50	100	100	15	5	5	5	30		
T9028246474	2016	MAAN PANCHAL	8	8	8	8	8	8	8	8	8	8	40	40	60	75	12	4	3	3.8	22.75		
Comprehensive Continous Evaluation (CCE) 15 Marks Distribution																							
Exam Seat No.	Roll No.	Name of Student											Cumulative Sum				15 Marks Distribution				Marks obtained out		
			Unit 1		Unit 2		Unit 3		Unit 4		Field Activity		Quiz		Internal Test		Field Activity		Internal Test				
			Field Activity	Quiz	Field Activity	Quiz	Field Activity	Quiz	Field Activity	Quiz													
			A	B	C	D	E	F	G	H			SUM(A+C+E+G)	SUM(B+D+F+H)			P	Q	R	S		SUM(P:Q)	
			10	10	10	10	10	10	10			40	40	100		5	5	5		15			
T9970160753	2020	AMOGH SHINDE	8	8	8	8	8	8	8			32	32	60		4	4	3		11			

Figure 1 Template Comprehensive Continuous Evaluation (CCE) [Click here to download](#) .

Field Activities / Home Assignments

Field activities and home assignments are essential components of experiential learning. Under this head, course projects, industrial visits, and guest lectures are to be incorporated. For each unit, one such activity should be designed and executed to reinforce theoretical learning through practical exposure.

1. Course Projects

Course Projects should be framed based on real-world problems relevant to the subject. Each course project must be communicated through one of the following modes. It is recommended to complete all the communication modes across different course projects:

- **Poster Presentation**
- **PowerPoint Presentation**
- **Model Making**
- **Field or Survey Report with Oral Presentation** (e.g., case study)
- **Submission of Digital Content (e.g. Video Summary)**

To evaluate these field activities, **assessment rubrics** should be designed. The rubrics should include criteria such as clarity, innovation, subject relevance, presentation skills, and technical content.

Note: Part of work of any co-curricular activities (relevant to subject contents) like national level project competitions, club activities, paper presentations, startup activities can be accepted as a course projects.

2. Industrial Visit

An industrial visit should be planned in alignment with the subject's scope and should particularly address advancements in the respective field. The purpose is to provide students exposure to actual engineering practices and systems.

Assessment of industrial visits should be carried out using any of the following tools:

- Quiz (based on the visit)
- Interactive video or oral discussion
- Submission of a detailed visit report

3. Guest Lectures

Guest lectures should be relevant to the course and highlight advanced topics or recent trends in the field. Subject experts from academia or industry may be invited.

Assessment methods for guest lectures may include:

- Quiz conducted post-lecture
- Attendance monitoring
- Evaluation of attentiveness and participation

Rubrics can be developed, if possible, to objectively assess student involvement in guest lectures.

4. Quiz

Unit-wise quizzes should be planned and can be conducted either **online** (via LMS, Google Forms) or

offline. Each quiz should include a **pool of 20 questions**, from which **students are required to attempt any 10**. The quizzes should be diversified across the following question types:

- Simple Multiple Choice Questions (MCQs)
- Numerical MCQs
- Image-based Questions
- Match the Following
- Fill in the Blanks
- Drag and Drop (using images or words)

This variety ensures the assessment caters to different cognitive skills and learning styles.

5. Internal Tests

Two major internal tests should be conducted as follows:

1. **Midterm Examination:** This should cover **Unit I and Unit II**, and should include questions targeting **Bloom's Taxonomy Levels 2, 3, and 4** (UNDERSTAND, APPLY, and ANALYZE).
2. **End term Examination:** This should cover the **remaining units** and should also include questions mapped to **BL Levels 2, 3, and 4**.

➤ CCE MODE 2:

Comprehensive Continuous Evaluation scheme in this mode for a theory subject of 30 marks with the specified parameters, the allocation of marks and the structure can be detailed as follows:

Sr. No.	Parameters	Marks	Coverage of Units
1	Unit Test	12 Marks	Unit 1 & Unit 2 (6 Marks/Unit)
2	Assignments/Case Study	12 Marks	Unit 3 & Unit 4 (6 Marks/Unit)
3	Seminar Presentation/ Open Book Test/ Quiz	06 Marks	Unit 5

6. CCE of 15 marks based on all the Units of course syllabus to be scheduled and conducted at institute level. To design a CCE scheme for a theory subject of 15 marks with the specified parameters, the allocation of marks and the structure can be detailed as follows:

Sr. No.	Parameters	Marks	Coverage of Units
1	Unit Test	10 Marks	Unit 1 & Unit 2 (5 Marks/Unit)
2	Seminar Presentation/ Open Book Test/ Assignments/ Case Studies	05 Marks	Unit 3 & Unit 4

• Unit Test–

Format: Questions designed as per Bloom's Taxonomy guidelines to assess various cognitive levels

(Remember, Understand, Apply, Analyze, Evaluate, and Create).

Implementation: Schedule the test after completing Units 1 and 2. Ensure the question paper is balanced and covers key concepts and applications.

- **Sample Question Distribution–**

1. Remembering (2 Marks): Define key terms related to [Topic from Units 1 and 2].
2. Understanding (2 Marks): Explain the principle of [Concept] in [Context].
3. Applying (2 Marks): Demonstrate how [Concept] can be used in [Scenario]
4. Analyzing (3 Marks): Compare & contrast [Two related concepts] from Units 1 and 2
5. Evaluating (3 Marks): Evaluate the effectiveness of [Theory/Model] in [Situation].

- **Assignments / Case Study:** Students should submit one assignment or one Case Study Report based on Unit 3 and one assignment or one Case Study Report based on Unit 4.

- **Format:** Problem-solving tasks, theoretical questions, practical exercises, or case studies that require in-depth analysis and application of concepts.

- **Implementation:** Distribute the assignments or case study after covering Units 3 and 4. Provide clear guidelines and a rubric for evaluation.

- **Seminar Presentation: –**

- Format:* Oral presentation on a topic from Unit 5, followed by a Q&A session. –

- Deliverables:* Presentation slides, a summary report in 2 to 3 pages, and performance during the presentation

- Implementation:* Schedule the seminar presentations towards the end of the course. Provide students with ample time to prepare and offer guidance on presentation skills.

- **Open Book Test: –**

- Format:** Analytical and application-based questions to assess depth of understanding. –

- Implementation:** Schedule the open book test towards the end of the course, ensuring it covers critical aspects of Unit 5

- **Quiz: –**

- Format:* Quizzes can help your students practice existing knowledge while stimulating interest in learning about new topic in that course. You can set your quizzes to be completed individually or in small groups.

- Implementation:* Online tools and software can be used create quiz. Each quiz is made up of a variety of question types including multiple choice, missing words, true or false etc

- **Example Timeline for conducting CCE: –**

1. Weeks 1-4 : Cover Units 1 and 2
2. Week 5 : Conduct Unit Test (12 marks)
3. Weeks 6-8 : Cover Units 3 and 4

4. Week 9 : Distribute and collect Assignments / Case Study (12 marks)
5. Weeks 10-12 : Cover Unit 5
6. Week 13 : Conduct Seminar Presentations or Open Book Test or Quiz (6 marks)

• **Evaluation and Feedback:** –

1. Unit Test: Evaluate promptly and provide constructive feedback on strengths and areas for improvement.
2. Assignments / Case Study: Assess the quality of submissions based on the provided rubric. Offer feedback to help students understand their performance.
3. Seminar Presentation: Evaluate based on content, delivery, and engagement during the Q&A session. Provide feedback on presentation skills and comprehension of the topic.
4. Open Book Test: Evaluate based on the depth of analysis and application of concepts. Provide feedback on critical thinking and problem-solving skills.

B) End-Semester Examination (ESE)

Detailed Scheme for 70 Marks: Unit-Wise Allocation (14 Marks per Unit): Each unit will have a combination of questions designed to assess different cognitive levels. By following this scheme, you can ensure a comprehensive and fair assessment of students' understanding and application of the course material, adhering to Bloom's Taxonomy guidelines for cognitive skills evaluation.

Detailed Scheme for 35 Marks: Unit-Wise Allocation (08 Marks for Unit 1 , 09 Marks for Unit 2, Unit 3 and Unit 4) : Each unit will have a combination of questions designed to assess different cognitive levels. By following this scheme, you can ensure a comprehensive and fair assessment of students' understanding and application of the course material, adhering to Bloom's Taxonomy guidelines for cognitive skills evaluation. The following structure is to be followed for designing an ESE for a **theory subject of 70 marks** covering **all 5 units** of the syllabus, with **questions set as per Bloom's Taxonomy** guidelines and **14 marks allocated per unit**.

2. Balanced Coverage

Ensure balanced coverage of all units with questions that assess different **cognitive levels of Bloom's Taxonomy**:

- a) **Remembering:** Basic recall of facts and concepts.
- b) **Understanding:** Explanation of ideas or concepts.
- c) **Applying:** Use of information in new situations.
- d) **Analyzing:** Drawing connections among ideas.
- e) **Evaluating:** Justifying a decision or course of action.
- f) **Creating:** Producing new or original work (if applicable).

Curriculum Structure - Semester V

NEP 2020 Compliant Curriculum Structure

Savitribai Phule Pune University															
Board of Studies - Mechanical and Automobile Engineering															
Undergraduate Program – Third Year Mechanical Engineering (2024 pattern)															
Course Code	Course Type	Course Name	Teaching Scheme (Hrs./week)			Examination Scheme and Marks						Credits			
			L	T	P	*CCE	ESE	TW	OR	PR	TOTAL	L	T	P	TOTAL
Semester –V															
PCC301-MEC	Major Course	Heat & Mass Transfer	4			30	70				100	4			4
PCC302-MEC	Major Course	Theory of Machines	3			30	70				100	3			3
PCC303-MEC	Major Course	Design of Machine Elements	4			30	70				100	4			4
PCC304-MEC	Major Course	Manufacturing Process-II	3			30	70				100	3			3
PCC305-MEC	Major Course	Metrology and Quality Control	3			30	70				100	3			3
PCC306-MEC	Major Course	Industrial Fluid Power Lab			2					50	50			1	1
MDM321-MEC	Multidisciplinary Course	Industrial Robotics			2					50	50			1	1
MDM 322-MEC	Multidisciplinary Course	Smart Mechatronics Systems			2					50	50			1	1
	Open Elective	OE is to be chosen compulsorily from faculty other than that of the Major Discipline.	2			15	35				50	2		0	2
Total			19	0	06	165	385	0	0	150	700	19	0	3	22

*CCE: Comprehensive Continuous Evaluation

Important Note: Min.1 to Max.2 hrs. per batch of (20-25 students) to be assigned for *CCE and to be considered in teaching load of concerned faculty. (Only applicable for Mode -1)

Curriculum Structure - Semester VI

NEP 2020 Compliant Curriculum Structure

Savitribai Phule Pune University

Board of Studies - Mechanical and Automobile Engineering

Undergraduate Program – Third Year Mechanical Engineering (2024 pattern)

Course Code	Course Type	Course Name	Teaching Scheme (Hrs./week)			Examination Scheme and Marks						Credits				
			L	T	P	CCE	ESE	TW	OR	PR	TOTAL	L	T	P	TOTAL	
Semester -VI																
PCC351-MEC	Major Course	Computer Integrated Manufacturing	4			30	70				100	4				4
PCC352-MEC	Major Course	Refrigeration & Air-conditioning	3			30	70				100	3				3
PCC353-MEC	Major Course	Transmission System Design	4			30	70				100	4				4
PCC354MEC	Major Course	Thermal and Fluid Engineering Lab-II			4					50	50			2		2
PEC 361-MEC	Program Elective -I	Students can pick the elective from basket	3			30	70				100	3				3
PEC 362-MEC		Students can pick the elective from basket	3			30	70				100	3				3
MDM 371-MEC	Multidisciplinary Course	Digital Manufacturing Laboratory			2					50	50			1		1
MDM 372-MEC	Multidisciplinary Course	Engineering Simulation and Analysis			2					50	50			1		1
VSE 381	Vocational Skill Course	Solar Technology and Maintenance			2	50					50	1		1		1
Total			17	0	10	200	350	0	0	150	700	18	0	5		22

Program Elective-I

PEC 361A-MEC	Industrial Tribology	PEC 362A-MEC	Product Design and Development
PEC 361B-MEC	Design of Heat Transfer Equipment's	PEC 362B-MEC	Smart Thermal Engineering systems
PEC 361C-MEC	Hydraulic and Pneumatics	PEC 362C-MEC	Compressible flow
PEC 361D-MEC	Industrial Engineering	PEC 362D-MEC	Advanced Machining Processes

***CCE: Comprehensive Continuous Evaluation**

Note: Students can opt for Open Electives offered by different faculties such as Arts, Science, Commerce, Management, Humanities, or Inter-Disciplinary Studies.